Separation between Second Price Auctions with Personalized Reserves and the Revenue Optimal Auction

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Abstract

When buyer valuations are drawn IID from a known regular distribution, a second price auction with a symmetric reserve price is the revenue-optimal single-item auction. When this distribution is irregular, we provide the first separation result showing that a second price auction with reserves earns at most 0.778 times the revenue of Myerson's optimal auction, even when the reserves can be asymmetric. Since the lower bound is 0.745 for i.i.d. buyers, our result is nearly tight.

Keywords: Bayesian mechanism design; single-item auctions; suboptimal mechanisms; second price auctions; sequential posted pricing; approximation ratios

1 Introduction

Approximating the optimal auction's expected revenue MyeRev by simple mechanisms has received a lot of attention in the algorithmic game theory literature, especially in the last decade. The seminal result of Myerson [10] describes the optimal auction in a single agent n-items setting when buyer valuations are drawn from a product distribution. When the buyer valuation distributions are independent and identical (i.i.d.) and satisfy a technical regularity condition, Myerson's optimal auction coincides with the second price (Vickrey's) auction with an anonymous reserve price (ASP). When the setting gets more complex, either because the distributions are not identical or because the distributions don't satisfy the regularity condition, or both, ASP ceases to be the optimal auction. Bounds on the suboptimality are known in many cases:

- when the distributions are non-identical (independent) and irregular, [1] show that ASP cannot get more than a $\frac{1}{n}$ fraction of MyeRev in general, and it is straightforward to show that this is tight;
- when the distributions are non-identical (independent) and regular, [7] show that ASP cannot get more than a $\frac{1}{2}$ fraction of MyeRev in general, [1] show that ASP can achieve a $\frac{1}{e}$ fraction of MyeRev in general, and recently [9] show that ASP cannot get more than a $\frac{1}{2.15}$ fraction of MyeRev in general;
- when the distributions are identical (independent) and irregular, the pricing problem can be reduced to the prophet inequalities problem on ironed virtual values, for which a $\frac{1}{2}$ approximation via a uniform threshold is known [12]. Since the distributions are identical, the uniform threshold in the ironed virtual value space translates into a uniform threshold in the value

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space, thus showing that ASP can achieve a $\frac{1}{2}$ approximation as well. This approximation factor of ASP is tight, as shown in [6], although we will also provide a self-contained proof here as a corollary.

On the contrary, *no upper bounds are known* on the fraction of MyeRev obtainable by a second price auction with personalized reserve prices. When the reserve prices are personalized, it matters whether the bidders are ranked after eliminating those below the reserve (eager, "ESP") or before eliminating those below the reserve (lazy, "LSP"). In the eager version, we first eliminate bidders below their respective reserves, and then charge the highest surviving bidder (if any) the maximum of his own reserve price and the second highest surviving bid. In the lazy version, we first decide the potential winner to be the highest bidder, eliminate him if he doesn't survive his reserve price (in which case we allocate the good to nobody), and then if he survives allocate the good to him at a price of maximum of his reserve price and the second highest bid. [11] show that for general correlated distributions, the optimal revenue obtainable from ESP and LSP are incomparable (i.e. either could be higher on a specific instance), but are within a factor of two. When the distributions are independent [11] show that the optimal ESP's revenue dominates the optimal LSP's revenue. Therefore from the perspective of upper bounding the fraction of MyeRev that is obtainable, any result for ESP implies a result for LSP. ESP's are also more widely used, and are a more natural auction to run—for instance, when there is at least one buyer that survives their reserve price, ESP always allocates the item, whereas LSP could result in no allocation.

Our result. Our main result (Theorem 3.4) is an upper bound on the maximum fraction of MyeRev that ESP can obtain. We show that in the n buyers single item setting, even when the buyer values are drawn i.i.d. (but not from a regular distribution), the fraction of MyeRev that ESP can extract in general is at most 0.778. The i.i.d. irregular setting is the simplest non-trivial setting to consider since, as discussed earlier, the i.i.d. regular setting results in a 1 approximation from even a second price auction with anonymous reserve price. We emphasize that no separation between the revenues of ESP and MyeRev was previously known, even for non-identical irregular valuations.

Corollary. Our construction can also be modified to show that ASP cannot get more than $\frac{1}{2}$ of MyeRev for i.i.d. irregular valuations, thus providing an alternate proof (to [6]) for the fact that the $\frac{1}{2}$ approximation factor for ASP is tight.

Near tightness of ESP result. A mechanism very related to ESP is the sequential posted price mechanism (SPM) where the seller computes one price per buyer, and then approaches the buyers in the descending order of posted prices until the item is sold. In general single-parameter environments with matroid feasibility constraints (this includes the single item setting we study as a special case), [3] showed the the revenue of an ESP that uses the posted prices of an SPM as its personalized reserve prices is at least as high as the SPM's revenue. This, combined with the recent result of [4] showing that SPM can achieve a 0.745 fraction of MyeRev in the i.i.d. setting with possibly irregular distributions, shows that ESP can also achieve at least a 0.745 approximation factor in the i.i.d. setting. Thus, our upper bound of 0.778 on ESP's approximation factor in the i.i.d. setting is quite close to this lower bound of 0.745. In the case of SPM both the upper and lower bound are known to be 0.745: the upper bound of 0.745 for SPMs was established by [8] about 3.5 decades prior to the matching lower bound established by [4]

ESP vs SPM: the benefit of simultaneity. The primary advantage of ESP over SPM is that ESP considers all buyers simultaneously while SPM considers buyers sequentially. Thus, as [3] show, ESP should only earn better revenue by benefiting from this simultaneity. Our upper bound of 0.778 on the approximation factor of ESP shows that this benefit of simultaneity is quite small in general, given that SPMs themselves can achieve a 0.745 approximation. In a sense, our result shows that *discrimination* is more crucial for revenue maximization than *simultaneity*. Closing this gap further, or establishing a separation between SPM and ESP remains an interesting open problem.

2 Preliminaries

We consider the standard single item n buyer Bayesian Mechanism Design problem where each buyer's value is drawn independently from a commonly known distribution F. In a seminar paper in Bayesian Mechanism Design, Myerson [10] derived the format of the auction which maximizes (expected) revenue. In the special case where the distribution has a strictly increasing cdf F with a well defined density f and the distribution is regular (namely: $x - \frac{1-F(x)}{f(x)}$ is weakly increasing in x), Myerson described the optimal auction as follows:

1. Compute the virtual value $\phi(v_i)$ of each agent i's value v_i as: $\phi(v_i) = v_i - \frac{1-F(v_i)}{f(v_i)}$.
2. Allocate the item to the agent $i^* = \arg\max_i \phi(v_i)$ if $\phi(v_{i^*}) \geq 0$; charge that agent the amount $\max(\phi^{-1}(0), \max_{i \neq i^*} v_i)$; charge all other agents 0.

When the distribution F is not regular, the above auction format is not incentive-compatible, i.e., revealing their true value is not guaranteed to be in an agent's best interest. Myerson applied a fix for this case through the *ironing* procedure. Define $q := 1 - F(v)$ to be the quantile of an agent with value v. Consider the revenue curve $R(q) := q \cdot F^{-1}(1-q)$, defined as the revenue obtainable by selling an item to a single agent (whose value is drawn from F) at a price of $F^{-1}(1-q)$ (note that $F^{-1}(\cdot)$ is unique given that F is strictly increasing). The virtual value $\phi(v)$ can be seen to be $R'(q)$ and the regularity of F is equivalent to the concavity of the curve $R(q)$. When the distribution F is not regular, the ironing procedure can be described as follows (see [2]):

1. Let $\bar{R}(q)$ be the concave envelope of $R(q)$, i.e., the smallest valued concave function in $[0, 1]$ such that $\bar{R}(q) \geq R(q)$.
2. Define $\bar{\phi}(v) := \bar{R}'(q)$. The function $\bar{\phi}(\cdot)$ is the *ironed virtual value* function.
3. Allocate the item to the agent $i^* = \arg\max_i \bar{\phi}(v_i)$ if $\bar{\phi}(v_{i^*}) \geq 0$. If i^* is not unique, there are two possible implementations: either pick one of the ironed virtual value maximizing agents uniformly at random (the *randomized* implementation), or break ties lexicographically (the *deterministic* implementation). Charge the winning agent the area above their allocation curve and charge all other agents 0.

While the above description of the optimal auction is for continuous distributions with a density, it can be extended to distributions with finite support. There are two approaches to do this. Elkind [5] extends virtual values to discrete distributions using the probability mass function (the analogue of the density function for discrete distributions). In this paper, instead of extending virtual values to discrete distributions as done in [5], we first extend revenue curves to discrete distributions as follows: $R(q)$ is still defined to be $R(q) := qF^{-1}(1-q)$, where the non-unique

nature of $F^{-1}(1-q)$ is resolved by taking it to be the largest price that results in a sale probability of at least q. And then, the virtual value is taken to be $R'(q)$.

The result of the approach we take is that our virtual values will differ slightly from those of Elkind [5], but our *ironed* virtual values will always agree, and hence the optimal auction format (and the optimal expected revenue) will be identical to that derived in Elkind's work[1] [5] (since allocations and payments are always determined through ironed virtual values). We adopt this approach because in our example below (in Section 3), the discrete virtual values according to [5] are actually monotonic (see Figure 1), yet a second price auction with reserves is suboptimal — in particular, in our example the optimal auction does not belong to the "either-one-among-the-highest-bidders-win-or-nobody-wins" family. There are some bid profiles where the non-highest-bidder has to win (regardless of a deterministic or randomized implementation) in the optimal auction.

Myerson's theorem on revenue and ironed virtual surplus. The main theorem of [10] is that the *expected revenue of any truthful auction is upper bounded by the expected ironed virtual surplus* (namely the expected ironed virtual value of agents who were served). When the auction's probability of serving an agent remains a constant over any value range for which the ironed virtual value is a constant, the *expected revenue will equal the expected ironed virtual surplus.* This characterization of Myerson immediately implies that the revenue optimizing auction is the one that allocates the item to the agent with the highest non-negative ironed virtual surplus. Note that this auction format naturally obeys the condition that the allocation probability remains unchanged in an interval where the ironed virtual value is a constant.

Ironing in continuous vs discrete distributions. When the distribution is continuous with a non-zero density everywhere, there is no need for ironing when the virtual value function is weakly increasing. On the other hand, for a discrete distribution, even when the virtual values (defined as $R'(q)$) are weakly increasing, ironing is required (following our approach above). This is because the revenue curve, being discontinuous in the discrete distributions case, will never be concave on its own and hence requires taking a concave envelope.

3 Upper bound on ESP approximation factor

We provide an example that establishes the upper bound on ESP's approximation factor.

Example 3.1. There are n buyers, with $n \to \infty$. Each buyer's valuation is drawn i.i.d. from a discrete distribution F where

$$v \sim F = \begin{cases} n, & \text{w.p. } 1/n^2 \\ \alpha/\beta, & \text{w.p. } \beta/n - 1/n^2 \\ 0, & \text{w.p. } 1 - \beta/n \end{cases}$$

Here $\alpha > 1$, $\beta = \omega(\frac{1}{n})$ s.t. $\alpha < n\beta$, and α, β are to be optimized later to minimize the fraction of MyeRev earned by ESP. While α and β will turn out to be constants, our revenue characterizations below of MyeRev and ESP in terms of α and β will hold for super constant α, β as well. For

[1] We note here that the virtual values in Elkind's work [5] are spelled out only at the discrete values in the support of the distribution and not at the other possible bids in between the discrete points. One could extend virtual value to such points as follows: the virtual value defined at v_i can be declared to be the virtual value for all bids in $[v_i, v_{i+1})$ where v_{i+1} is the immediately larger value than v_i in the support of the distribution.

instance we set $\alpha = \beta = n$ to derive the alternate proof of tightness of $\frac{1}{2}$ approximation factor of ASP to MyeRev.

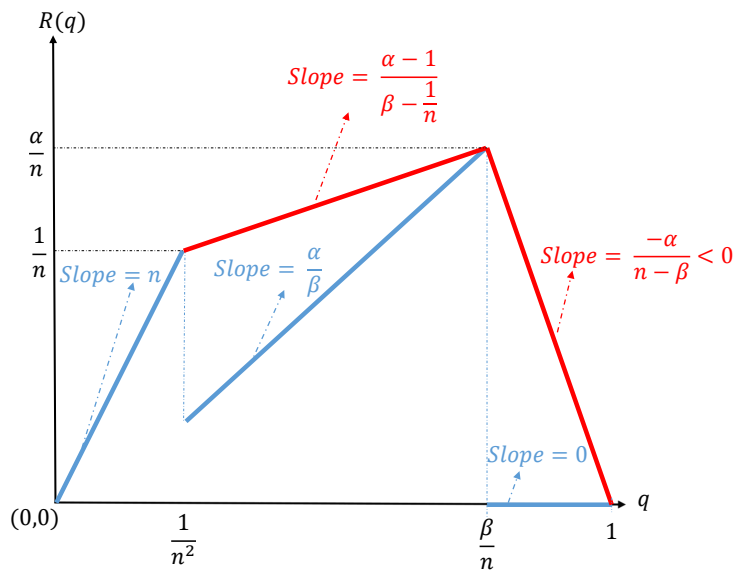

Figure 1: Revenue curve $R(q)$ (in solid blue) and its slope (i.e., virtual value); the concave envelope of the revenue curve consists of the first solid blue segment, and the following two solid red segments; the ironed virtual value is the slope of the concave envelope. For this distribution, the "virtual values" defined according to [5] are identical to our "ironed virtual values".

Lemma 3.2. *In a n buyers setting, when the buyer valuations are drawn i.i.d. from the distribution F described in Example 3.1, the expected revenue* MyeRev *of Myerson's optimal auction approaches*

$$1 + \frac{\alpha - 1}{\beta}(1 - e^{-\beta}).$$

Proof. We compute the revenue as the expected ironed virtual surplus (see Section 2 for Myerson's theorem on this). To compute the latter, consider the revenue curve $R(q) = qF^{-1}(1-q)$, where in this case $F^{-1}(1-q)$ denotes the maximum price at which the probability of getting a sale (from a single buyer with valuation drawn from F) is at least q. It can be computed that

$$R(q) = qF^{-1}(1-q) = \begin{cases} qn, & 0 < q \leq 1/n^2; \\ q\alpha/\beta, & 1/n^2 < q \leq \beta/n; \\ 0, & \beta/n < q \leq 1. \end{cases}$$

$$\phi(v) = \begin{cases} n, & v = n; \\ \alpha/\beta, & v = \alpha/\beta; \\ 0, & v = 0. \end{cases} \qquad \bar{\phi}(v) = \begin{cases} n, & v = n; \\ \frac{\alpha-1}{\beta-1/n}, & v = \alpha/\beta; \\ -\frac{\alpha}{n-\beta}, & v = 0. \end{cases}$$

The discontinuous revenue curve (see the solid blue lines in Figure 1) is clearly not concave. The ironed revenue curve, which is the concave envelope of the revenue curve, consists of the first solid blue segment and the next two solid red segments in Figure 1. The slope of three line segments in the ironed revenue curve are the ironed virtual values corresponding to the three value realizations.

The complete description of Myerson's optimal mechanism here is as follows:

- any buyer who bids below $\frac{\alpha}{\beta}$ is never allocated, and always pays 0;
- if there are no buyers with bid n or larger: allocate the item to a uniformly random buyer among all buyers with bids in $[\frac{\alpha}{\beta}, n)$ and charge him $\frac{\alpha}{\beta}$;
- if there is exactly one buyer with bid of n or larger, that buyer gets the item with probability 1, and he pays $n - (n - \frac{\alpha}{\beta}) \cdot 1/(1 + \text{num of buyers with bids in } [\frac{\alpha}{\beta}, n))$ which is the area above the allocation curve of that buyer;
- if there is more than one buyer with bid of n or larger, allocate the item to a uniformly random buyer among all buyers with bids in $[n, \infty)$ and charge him n.

The expected revenue of Myerson's optimal mechanism MyeRev equals the expected ironed virtual surplus, which can be computed as follows. With probability $1 - (1 - 1/n^2)^n$, at least one buyer has virtual valuation n. With probability $(1 - 1/n^2)^n - (1 - \beta/n)^n$, no buyer has virtual valuation n but at least one buyer has ironed virtual valuation $\frac{\alpha-1}{\beta-1/n}$. Therefore, the expected ironed virtual surplus, as the number of buyers approaches ∞, is

$$\lim_{n\to\infty} \left(n\Big(1 - (1 - \frac{1}{n^2})^n\Big) + \frac{\alpha - 1}{\beta - 1/n}\Big((1 - \frac{1}{n^2})^n - (1 - \frac{\beta}{n})^n\Big)\right) = 1 + \frac{\alpha - 1}{\beta}(1 - e^{-\beta}),$$

completing the proof. □

Lemma 3.3. *In a n buyers setting, when the buyer valuations are drawn i.i.d. from the distribution F described in Example 3.1, the expected revenue* EspRev *of* ESP *as $n \to \infty$ is at most*

$$1 + \frac{\alpha - \ln \alpha - 1}{\beta}.$$

Proof. Consider any revenue-maximizing ESP auction. Since the only possible non-zero valuations are n and α/β, it does not lose generality to assume that all reserve prices are set to either n or α/β. Let $z \in \{0, \frac{1}{n}, \frac{2}{n}, \ldots, 1\}$ denote the fraction of buyers whose reserve is set to n, so that the number of buyers with reserves n and α/β are zn and $(1 - z)n$, respectively.

With probability $1 - (1 - \frac{1}{n^2})^{zn}$, at least one of the buyers with reserve n clears their reserve price, and the ESP auction earns the maximum possible revenue of n. Otherwise, all zn buyers with reserve n are eliminated from the auction and we consider the other buyers.

For the other $(1-z)n$ buyers with reserve α/β, at least one of them will clear their reserve with probability $1 - (1 - \frac{\beta}{n})^{(1-z)n}$, in which case the ESP auction is guaranteed to earn at least α/β. In this case, there is also the possibility of earning the higher revenue of n, if at least two buyers report a valuation of n. However, the probability of this event is at most $\binom{n}{2}$ times the probability that a *given* pair of buyers both report n, by the union bound. Therefore, the expected revenue collected from this event is at most $n\binom{n}{2}(\frac{1}{n^2})^2 = O(\frac{1}{n})$, and can be ignored as $n \to \infty$.

Collecting the above terms, for any n and $z \in \{0, \frac{1}{n}, \frac{2}{n}, \ldots, 1\}$, the expected revenue of the corresponding ESP auction is

$$\begin{aligned}
& n\Big(1-(1-\frac{1}{n^2})^{zn}\Big) + \frac{\alpha}{\beta}(1-\frac{1}{n^2})^{zn}\Big(1-(1-\frac{\beta}{n})^{(1-z)n}\Big) \\
\leq\ & n\Big(1-(1-\binom{zn}{1}\frac{1}{n^2})\Big) + \frac{\alpha}{\beta}(1)\Big(1-e^{-\beta(1-z)}\Big) \\
=\ & z + \frac{\alpha}{\beta}\Big(1-e^{-\beta(1-z)}\Big) + O(\frac{1}{n}) \qquad (1)
\end{aligned}$$

where we used the binomial theorem in the first inequality. The maximum of the concave function $z + \frac{\alpha}{\beta}(1-e^{-\beta(1-z)})$ over $z \in [0,1]$ is attained at $z = 1 - \frac{\ln\alpha}{\beta}$, and equals $1 - \frac{\ln\alpha}{\beta} + \frac{\alpha}{\beta}(1-\frac{1}{\alpha})$, completing the proof. □

Theorem 3.4. *Let* $\textsc{MyeRev}(n, \mathcal{D})$ *and* $\textsc{EspRev}(n, \mathcal{D})$ *denote respectively the revenue of the optimal auction (Myerson's auction) and the second price auction with personalized reserves in a single-item setting with* n *buyers whose values are drawn i.i.d from* $\mathcal{D}$*. Then:*

$$\inf_{n,\mathcal{D}} \frac{\textsc{EspRev}(n, \mathcal{D})}{\textsc{MyeRev}(n, \mathcal{D})} < 0.778.$$

Proof. Consider Example 3.1. By Lemmas 3.2 and 3.3, the fraction of Myerson's optimal revenue earned by an ESP auction in this example (as $n \to \infty$) is at most

$$\frac{\beta + \alpha - 1 - \ln\alpha}{\beta + (\alpha-1)(1-e^{-\beta})}, \qquad (2)$$

for any values of $\alpha > 1$ and $\beta > 0$. By numerically minimizing over such values of α and β, it can be checked that when $\alpha = 2.91$ and $\beta = 1.89$, the value of (2) is less than 0.778, completing the proof. □

3.1 Upper bound on ASP approximation factor

Consider the distribution from Example 3.1 with $\alpha = \beta = n$ and let $n \to \infty$. By Lemma 3.2, the optimal revenue of Myerson is

$$\lim_{n\to\infty}\left(n\Big(1-(1-\frac{1}{n^2})^n\Big) + \frac{n-1}{n-1/n}\Big((1-\frac{1}{n^2})^n - (1-\frac{n}{n})^n\Big)\right) = 2.$$

On the other hand, ESP must set all of the reserves to either n or 1. By the analysis in Lemma 3.3, this is upper-bounded by expression (1) with $z = 0$ or $z = 1$, i.e. the revenue of ASP is at most

$$\max\left\{1 - e^{-n} + O(\frac{1}{n}), 1 + O(\frac{1}{n})\right\}$$

which approaches 1 as $n \to \infty$. This completes the proof.

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